

Ryan Pham

Fountain Valley, CA | ryandphambusiness@gmail.com | [GitHub](#) | [Portfolio](#)

EDUCATION

California State Polytechnic University, Pomona

Expected Graduation Date: December 2026

B.S. Computer Science

GPA: 3.92

- Organizations: Google Developer Student Clubs, Computer Science Society, Data Science & AI Club, Bronco Space Lab
- Honors: President's List (2x), Dean's List (4x)

SKILLS

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL

Technologies: React.js, FastAPI, Node.js, Git, PostgreSQL, MongoDB, SQLite, PyTorch, F Prime (F')

EXPERIENCE

CADENCE-SWANS CubeSat (AFRL University Nanosatellite Program)

Pomona, CA

Flight Software Engineer

May 2026 - Present

- Engineered the Real-Time Clock (RTC) Manager in C++ using NASA's F Prime (F') framework for a 12U CubeSat, defining ports, commands, telemetry channels, and interfaces for spacecraft timekeeping.
- Architecting the flight software topology by mapping F Prime components to planned spacecraft hardware and establishing interfaces between avionics, software components, and mission subsystems under UNP requirements.
- Configured the RTC Manager within the F Prime system topology, linking supporting components and preparing the design for future hardware integration, verification, and system-level testing.

CIC | PCUBED Data Science Summer Research Project

Fullerton, CA

Research Scholar

June 2023 - August 2023

- Trained machine learning models with Python and scikit-learn on 28K+ police records, engineering temporal, spatial, and demographic features to achieve 80% forecasting accuracy on New York City shooting incident data.
- Processed and standardized inconsistent real-world datasets through data cleaning, transformation, and feature preparation to improve data quality and experiment reproducibility.

Code Ninjas

Fountain Valley, CA

Programming Instructor

March 2023 - August 2024

- Instructed 100+ students in Python, JavaScript, Java, and C# through hands-on, project-based programming curricula spanning elementary through high school levels.
- Directed development camps using Unity/C#, Roblox Studio/Lua, and TinkerCAD, maintaining a 95% student retention rate through interactive technical instruction.

NASA L'SPACE Mission Concept Academy

Remote

Team Member

June 2023 - August 2023

- Collaborated within a 13-person interdisciplinary team to formulate a Mars mission concept for investigating crustal magnetic fields in the Southern Highlands region.
- Gathered and analyzed information on potential instruments for use, while managing project timelines, budgets, and resource allocation to ensure the feasibility and efficiency of the mission plan.

PROJECTS

UAV Drone Tracker | Python, YOLO, ByteTrack, OpenCV, NumPy

- Engineered a real-time multi-UAV computer vision pipeline using YOLO, ByteTrack, and OpenCV, associating detections across video frames with persistent track IDs and independent state histories for each UAV.
- Applied Kalman filtering and motion estimation to position and velocity states, smoothing noisy detections and maintaining continuously updated motion estimates for individual UAV tracks.
- Designed multi-horizon trajectory forecasting at 0.5, 1.0, and 2.0 seconds, projecting future UAV positions from filtered motion states and maintaining prediction histories for subsequent evaluation.
- Recorded per-UAV telemetry and motion histories by track ID, preserving time-series state data for real-time visualization, trajectory analysis, and prediction evaluation.

CineMatch | Python, FastAPI, React, OpenAI API, MCP, TMDB API

- Developed a full-stack agentic AI application spanning 3 services — React frontend, FastAPI backend, and MCP tool server — using REST APIs, SSE, and OpenAI tool calling for personalized movie and TV recommendations.
- Built a Model Context Protocol (MCP) server with 3 schema-validated tools routing 100% of TMDB access through a single layer; required per-title detail/trailer tool calls so recommendations use verified metadata (posters, runtime, ratings) instead of LLM hallucination.
- Reduced TMDB search/discover latency by ~50% via parallel async HTTP (httpx, asyncio.gather) across movie and TV endpoints; implemented an agent loop supporting up to 24 concurrent tool-call rounds with structured JSON output.

Buzzr | JavaScript, Node.js, SQLite, Twilio API, Discord.js

- Deployed a Node.js Discord application serving 100+ users across 2 sports communities, automating announcements and event-driven notification workflows.
- Integrated the Twilio API to process and deliver 10+ daily SMS alerts, extending server notifications to subscribed mobile users.